

Name: MODEL SOLUTIONS.

Student Number:

Computer you are using:

(for example: MECH1810 or ENDCAD229 – there is a label either on the monitor, or on the computer under the desk)

Please answer ALL questions.

This test is worth 25% of your overall class grade.
Each problem statement has the associated mark value stated.

Do not turn over this cover sheet to start until you are told to begin. Extra blank pages are given for each problem with what should be enough room to answer each question, especially if you use both sides. However, if you need extra paper, please just ask and it will be provided.

If you separate the pages, ensure that they are stapled together again when handed in.

There is a total of 45 marks (22 marks for Question 1 and 23 marks for Question 2)

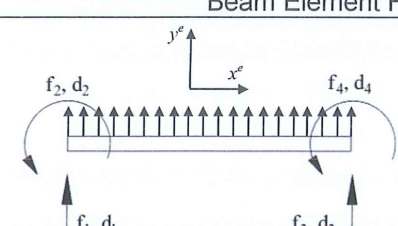
90 minutes total time allowed.

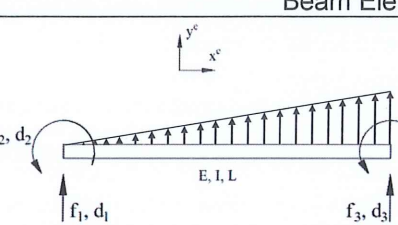
Circle or box all final answers! You may use pencil but make sure to highlight and/or ink final answers.

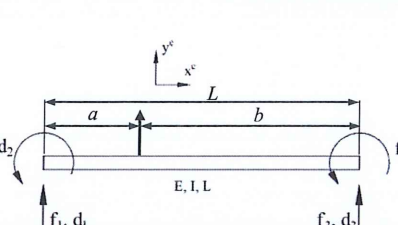
Label each part of each question and include appropriate units for all answers.

Some equations for equivalent nodal loading are presented on the next page for your reference.

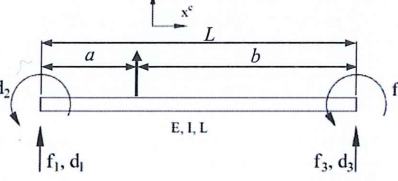
The equivalent nodal loading definitions (in element co-ordinates) are provided for reference.

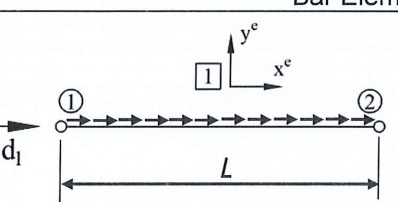
Equivalent nodal loading definition for a Uniformly Distributed Load (UDL)		
Beam Element Formulation	Frame Element Formulation	
	$f_{eq,UDL}^e = \begin{Bmatrix} \bar{w}L/2 \\ \bar{w}L^2/12 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix}$	$f_{eq,UDL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ \bar{w}L/2 \\ \bar{w}L^2/12 \\ 0 \\ \bar{w}L/2 \\ -\bar{w}L^2/12 \end{Bmatrix}$

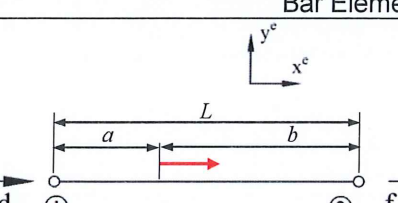
Equivalent nodal loading definition for a Linearly Varying Load (LVL)		
Beam Element Formulation	Frame Element Formulation	
	$f_{eq,LVL}^e = \begin{Bmatrix} 3\bar{w}L/20 \\ \bar{w}L^2/30 \\ 7\bar{w}L/20 \\ -\bar{w}L^2/20 \end{Bmatrix}$	$f_{eq,LVL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ 3\bar{w}L/20 \\ \bar{w}L^2/30 \\ 0 \\ 7\bar{w}L/20 \\ -\bar{w}L^2/20 \end{Bmatrix}$

Equivalent nodal loading definition for a transverse point load		
Beam Element Formulation	Frame Element Formulation	
	$f_{eq,PL}^e = \bar{w} \begin{Bmatrix} 1 - \frac{3a^2}{L^2} + \frac{2a^3}{L^3} \\ \frac{a^3}{L^2} - \frac{2a^2}{L} + a \\ \frac{3a^2}{L^2} - \frac{2a^3}{L^3} \\ \frac{a^3}{L^2} - \frac{a^2}{L} \end{Bmatrix}$	$f_{eq,PL}^e = \bar{w} \begin{Bmatrix} 0 \\ 1 - 3(a^2/L^2) + 2(a^3/L^3) \\ (a^3/L^2) - 2(a^2/L) + a \\ 0 \\ 3(a^2/L^2) - 2(a^3/L^3) \\ (a^3/L^2) - (a^2/L) \end{Bmatrix}$

Specific, equivalent nodal loading definition for a mid-span point load, where $a = L/2$

Beam Element Formulation	Frame Element Formulation	
	$f_{eq,PL}^e = \begin{Bmatrix} \bar{w}/2 \\ \bar{w}L/8 \\ \bar{w}/2 \\ -\bar{w}L/8 \end{Bmatrix}$	$f_{eq,PL\{6 \times 1\}}^e = \begin{Bmatrix} 0 \\ \bar{w}/2 \\ \bar{w}L/8 \\ 0 \\ \bar{w}/2 \\ -\bar{w}L/8 \end{Bmatrix}$

Equivalent nodal loading definition for a Distributed Axial Load		
Bar Element Formulation	Frame Element Formulation	
	$f_{eq,axial}^e = \bar{p} \begin{Bmatrix} L/2 \\ L/2 \end{Bmatrix}$	$f_{eq,axial}^e = \bar{p} \begin{Bmatrix} L/2 \\ 0 \\ 0 \\ L/2 \\ 0 \\ 0 \end{Bmatrix}$

Equivalent nodal loading definition for a Concentrated Axial Load		
Bar Element Formulation	Frame Element Formulation	
	$f_{eq,axial PL}^e = \bar{p} \begin{Bmatrix} 1 - a/L \\ a/L \end{Bmatrix}$	$f_{eq,axial PL}^e = \bar{p} \begin{Bmatrix} 1 - a/L \\ 0 \\ 0 \\ a/L \\ 0 \\ 0 \end{Bmatrix}$

Problem 1:

A steel structure is modelled as three frame elements, as shown on the next page. The support at the left edge of the structure is considered to be a fixed support and the support under the right end of element 1 is a roller support that can provide a vertical reaction force, but cannot provide any horizontal force or moment reaction. The structure itself is considered mass-less relative to the loads it carries.

A downwards uniformly distributed load with intensity of **25.0** kN/m is applied to the top of element 1.

An upwards concentrated load of **40.0** kN and a horizontal load of **20.0** kN acting to the right are applied at the top right-hand corner of the structure, as shown on the next page.

The element properties are given with the sketch on the next page.

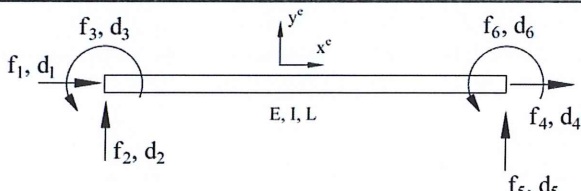
[22 Total Marks]

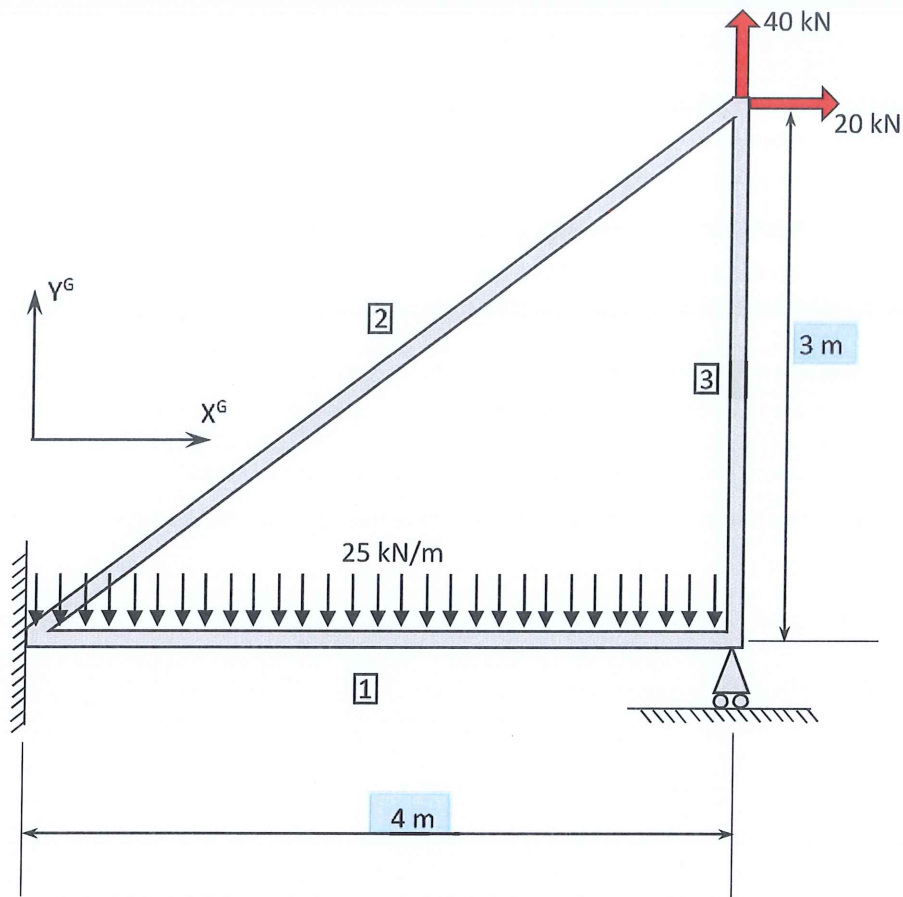
- Draw a sketch of the structure showing the global degrees of freedom **q**. **[2 Marks]**
- Draw a free-body diagram for each element, showing the element degrees of freedom in global co-ordinates as well as the orientation of your element co-ordinate system (including stating the transformation angle, α , for each element). **[2 Marks]**
- Write out the assembly matrices **A^e** for each element **[2 Marks]**
- Write out the element contributions to the global stiffness matrix (**KG1**, **KG2**, and **KG3**) of the structure. You do not need to write out the overall matrix **KG**. **[2 Marks]**
- Write out the element equivalent forcing terms for element 1 (**f^{1eq}**) to represent the Uniformly Distributed Load. Also present the overall global forcing vector, **Q**, that includes all loads.

Note: The formulae on the second page of this test may be a useful reference.

[3 Marks]

- Solve for the overall deflections of the system, **q**, and present your result. **[1 Marks]**
- Sketch (by hand) the original and deflected shape of the structure, indicating the direction and magnitude of all deflection components (displacements and rotations). The sketch does not need to be to scale, but each component of deflection must be labelled. **[3 Marks]**
- What are the reaction loads provided by each of the supports? Provide a sketch showing the magnitude and direction of all reaction components at both supports, parallel and perpendicular to the supports. **[3 Marks]**
- What is the maximum magnitude of the bending moment induced within element 2 and where is it located within the element? **[2 Marks]**
- Determine the horizontal (X^G) deflection at the mid-span of element 1 ($x = L/2$). The shape functions (applied to the frame element numbering system) are given below. Include your working. **[2 Marks]**

List of the shape functions for a frame element.			
$\psi_1(x) = \left(1 - \frac{x}{L}\right); \psi_2(x) = \frac{x}{L}$			
$N_1(x) = 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3}$	$N_2(x) = \frac{x^3}{L^2} - \frac{2x^2}{L} + x$	$N_3(x) = \frac{3x^2}{L^2} - \frac{2x^3}{L^3}$	$N_4(x) = \frac{x^3}{L^2} - \frac{x^2}{L}$
		$u(x) = \psi_1(x)d_1 + \psi_2(x)d_4$ $v(x) = N d^e = \begin{pmatrix} N_1(x) & N_2(x) & N_3(x) & N_4(x) \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$	



All elements are FRAME elements

Elastic Modulus: $E_1 = E_2 = E_3 = 200.0$ GPa

Second Moment of Area: $I_1 = I_2 = I_3 = 5.00 \times 10^{-6}$ m⁴

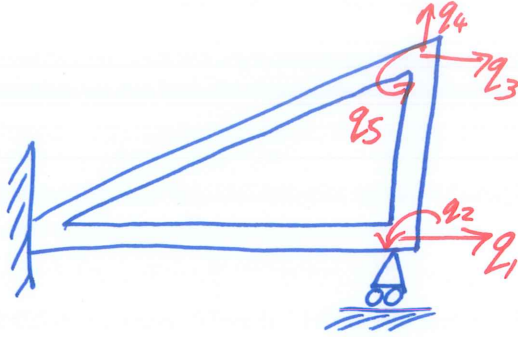
Cross-sectional Area: $A_1 = A_2 = A_3 = 2.00 \times 10^{-4}$ m²

Lengths: $L_1 = 4.00$ m

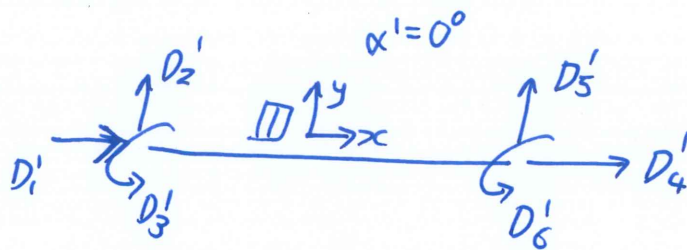
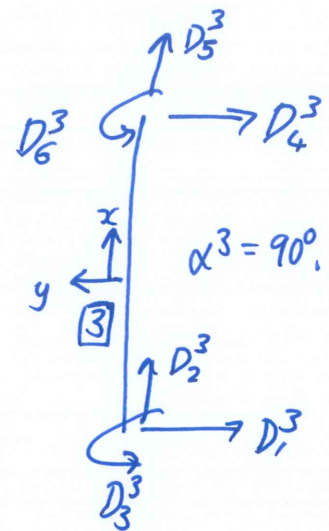
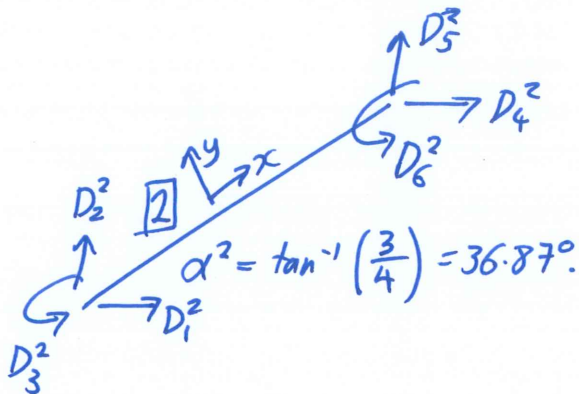
$L_2 = 5.00$ m

$L_3 = 3.00$ m

Q1a)



Q1b)



Q1c)

$$A^1 = \begin{matrix} & D_1^1 & D_2^1 & D_3^1 & D_4^1 & D_5^1 & D_6^1 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A^2 = \begin{matrix} & D_1^2 & D_2^2 & D_3^2 & D_4^2 & D_5^2 & D_6^2 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$A^3 = \begin{matrix} & D_1^3 & D_2^3 & D_3^3 & D_4^3 & D_5^3 & D_6^3 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Q(d)

$$k_G' = 1 \times 10^7 \times \begin{bmatrix} 1.0 & 0 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$k_G^2 = 1 \times 10^6 \times \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5.1546 & 3.7939 & 0.1440 \\ 0 & 0 & 3.7939 & 2.9414 & -0.1920 \\ 0 & 0 & 0.1440 & -0.1920 & 0.8000 \end{bmatrix}$$

$$k_G^3 = 1 \times 10^7 \times \begin{bmatrix} 0.0444 & -0.0667 & -0.0444 & -0.0000 & -0.0667 \\ -0.0667 & 0.1333 & 0.0667 & -0.0000 & 0.0667 \\ -0.0444 & 0.0667 & 0.0444 & 0.0000 & 0.0667 \\ -0.0000 & -0.0000 & 0.0000 & 1.3333 & -0.0000 \\ -0.0667 & 0.0667 & 0.0667 & -0.0000 & 0.1333 \end{bmatrix}$$

$$k_G = 1 \times 10^7 \times \begin{bmatrix} 1.0444 & -0.0667 & -0.0444 & -0.0000 & -0.0667 \\ -0.0667 & 0.2333 & 0.0667 & -0.0000 & 0.0667 \\ -0.0444 & 0.0667 & 0.5599 & 0.3794 & 0.0811 \\ -0.0000 & -0.0000 & 0.3794 & 1.6275 & -0.0192 \\ -0.0667 & 0.0667 & 0.0811 & -0.0192 & 0.2133 \end{bmatrix}$$

Q1 e):

$$f'_{eq, uoL} = \bar{w} \begin{Bmatrix} 0 \\ L/2 \\ (L)^2/12 \\ 0 \\ L/2 \\ (L)^2/12 \end{Bmatrix} = (-25,000 \text{ N/m}) \begin{Bmatrix} 0 \\ 4/2 \\ 4^2/12 \\ 0 \\ 4/2 \\ -4^2/12 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ -50,000 \text{ N} \\ -33,333 \text{ Nm} \\ 0 \text{ N} \\ -50,000 \text{ N} \\ +33,333 \text{ Nm} \end{Bmatrix}$$

due to $\alpha' = 0^\circ$, $\Lambda' = I_{6 \times 6}$ and $F'_{eq, uoL} = f'_{eq, uoL}$

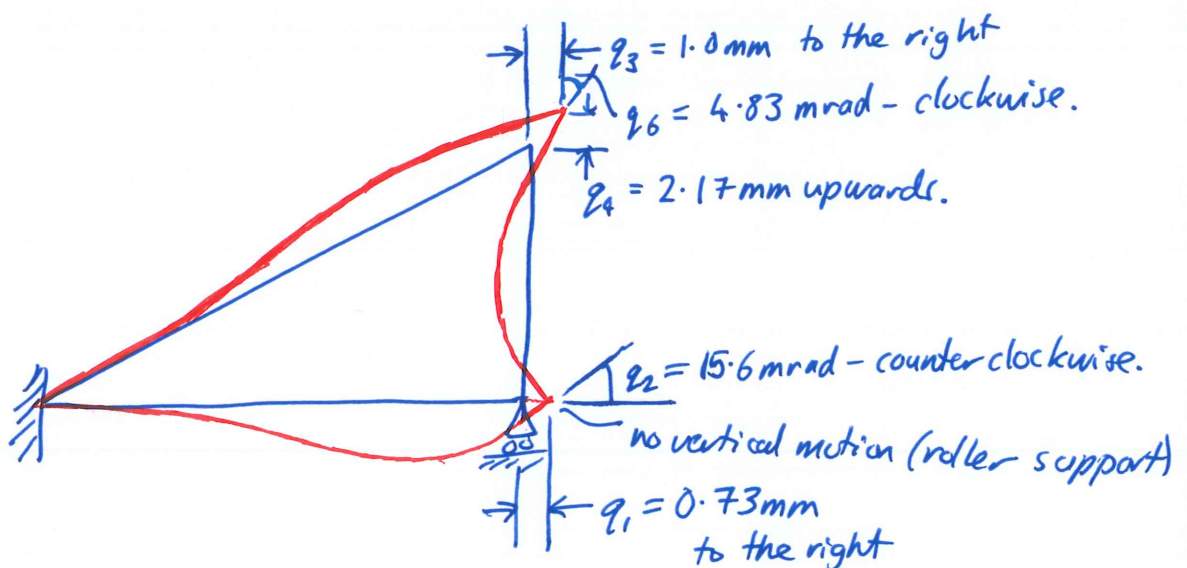
$$Q'_{eq, uoL} = A' F'_{eq, uoL} = \begin{Bmatrix} 0 \text{ N} \\ 33,333 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$$

$$Q_{total} = Q_{nodal} + Q'_{eq, uoL} = \begin{Bmatrix} 0 \text{ N} \\ 0 \text{ Nm} \\ 20,000 \text{ N} \\ 40,000 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix} + \begin{Bmatrix} 0 \text{ N} \\ 33,333 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ 33,333 \text{ Nm} \\ 20,000 \text{ N} \\ 40,000 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$$

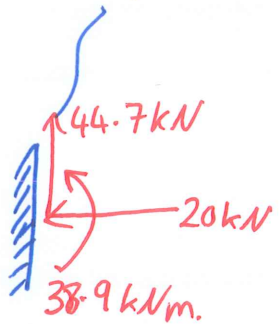
Q1 f):

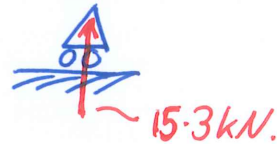
$$q = KG \backslash Q_{total} = \begin{Bmatrix} 0.0007 \text{ m} \\ 0.0156 \text{ rad} \\ 0.0010 \text{ m} \\ 0.0022 \text{ m} \\ -0.0048 \text{ rad} \end{Bmatrix} = \begin{Bmatrix} 0.7294 \text{ mm} \\ 15.5864 \text{ mrad} \\ 1.0054 \text{ mm} \\ 2.1664 \text{ mm} \\ -4.8300 \text{ mrad} \end{Bmatrix}$$

Q1 g):



Q1h):

$$F'(1:3) + F^2(1:3) + (-F_{eq, uol}(1:3)) = \begin{Bmatrix} -7.2939 \text{ kN} \\ 5.8451 \text{ kN} \\ 7.7934 \text{ kNm} \end{Bmatrix} + \begin{Bmatrix} -12.7061 \text{ kN} \\ -11.1142 \text{ kN} \\ -2.2032 \text{ kNm} \end{Bmatrix} + \begin{Bmatrix} 0 \text{ kN} \\ 50 \text{ kN} \\ 33.3 \text{ kNm} \end{Bmatrix} = \begin{Bmatrix} -20 \text{ kN} \\ 44.7309 \text{ kN} \\ 38.9236 \text{ kNm} \end{Bmatrix}$$




$$F'(4:6) + F^3(1:3) + (-F_{eq, uol}(4:6)) = \begin{Bmatrix} 7.2939 \text{ kN} \\ -5.8451 \text{ kN} \\ 15.5869 \text{ kNm} \end{Bmatrix} + \begin{Bmatrix} -7.2939 \text{ kN} \\ -28.8858 \text{ kN} \\ 17.7465 \text{ kNm} \end{Bmatrix} + \begin{Bmatrix} 0 \text{ kN} \\ 50 \text{ kN} \\ -33.3 \text{ kNm} \end{Bmatrix} = \begin{Bmatrix} 0 \text{ kN} \\ 15.2691 \text{ kN} \\ 0 \text{ kNm} \end{Bmatrix}$$

Q1i):

$$F^2 = \begin{Bmatrix} -12.7061 \text{ kN} \\ -11.1142 \text{ kN} \\ -2.2032 \text{ kNm} \\ 12.7061 \text{ kN} \\ 11.1142 \text{ kN} \\ -4.1352 \text{ kNm} \end{Bmatrix}$$

Based upon the assumption of a constant shear force within the element, the bending moment will vary linearly from -2.2032 kNm to -4.1352 kNm .

⇒ The magnitude of the maximum bending moment is 4.14 kNm and occurs at the top of element [2] (which is node ② based upon our chosen element orientation defined in Q1b).

Q1j)

$$d' = \begin{Bmatrix} 0\text{mm} \\ 0\text{mm} \\ 0\text{mrad} \\ 0.7294\text{mm} \\ 0\text{mm} \\ 15.5889\text{mrad} \end{Bmatrix}$$

As we are only asked for the X^G translation and this element has a 0° element transformation angle, we only need $u(x)$ and not $v(x)$.

$$u(x) = \psi_1(x) d'_1 + \psi_2(x) d'_4$$

$$= \left(\frac{x}{L}\right) d'_4$$

$$u(x = \frac{L'}{2}) = \left(\frac{2}{4}\right) (0.7294\text{mm}) = 0.3647\text{mm}.$$

and we know that $u(x)$ for this element acts in the δX^G direction, so X^G at $x = \frac{L'}{2}$ is 0.365mm .

Alternatively:

$$\delta X^G = u(x) \cos(\alpha') - v(x) \sin(\alpha')$$

$$= (0.3647)(1.0)$$

$$= 0.3647\text{mm}.$$

$\rightarrow 0$ as $\alpha' = 0^\circ$.

Problem 2: The diagram on the next page shows an architecturally-designed structure that supports a shelter with an overhead roof. Snow has fallen onto the roof and creates a 1.0 kPa pressure on the roof that acts vertically downwards due to gravity.

The structure is made up of three frame elements and has two fixed supports – one at the left side of the structure and one at the bottom of the structure.

Assume that all elements are made of steel, with the same cross-sectional area and second moment of area, with values given on the next page. The length of the elements differs as given on the next page, and the structure itself is considered mass-less relative to the loads it carries. **[23 Total Marks]**

- Draw a sketch of the structure showing the global degrees of freedom q for the model. **[2 Marks]**
- Draw a free-body diagram for each element, showing the element degrees of freedom in global co-ordinates as well as the orientation of your element co-ordinate system (including stating the transformation angle, α , for each element). **[2 Marks]**
- Write out the assembly matrices A^e for each element **[2 Marks]**
- Assume that the frame being modelled supports a section of roof (the distance into the page) of **5.00 m**.

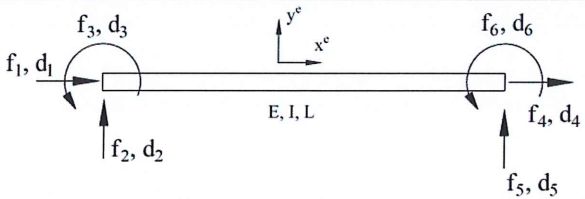
Write out the element equivalent forcing terms for elements 1 and 2 (f_{eq}^1 and f_{eq}^2) to represent the snow loading. For element 2 (f_{eq}^2) present both the axial and shear components of the distributed load (these will be a 6x1 vectors). Present the final overall global forcing vector, Q .

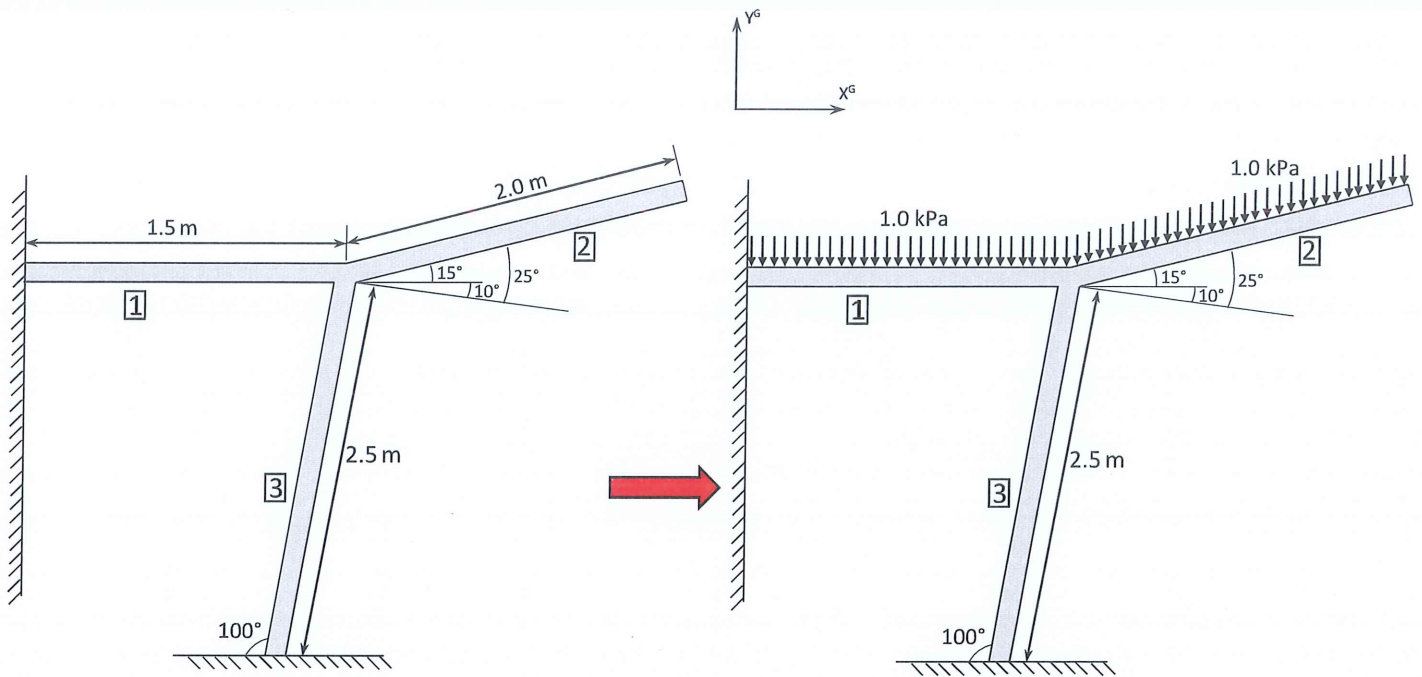
Note: The formulae on the second page of this test may be a useful reference. If you are unsure how to do this step, just apply the snow load as a pure shear load on both elements and continue with the question.

[4 Marks]

If you cannot calculate the intensity of the uniformly distributed load, use a magnitude of $\bar{w} = 6.0$ kN/m.

- Solve for the overall deflections of the system, q , and present your result. **[1 Mark]**
- Sketch (by hand) the original and deflected shape of the structure, indicating the direction and magnitude of all deflection components (displacements and rotations). The sketch does not need to be to scale, but each component of deflection must be labelled (including appropriate units). **[3 Marks]**
- Provide the support reaction forces and moments at the fixed supports, parallel and perpendicular to the supports. Provide a sketch of these reaction loads, showing the magnitude and direction of all reaction components. **[3 Marks]**
- Present the element displacement vector (in element co-ordinates) for element 3. Determine the axial normal strain in element 3 (from axial loads only, ignoring any bending effects), and indicate whether this strain is tensile or compressive. **[2 Marks]**
- Based upon your element displacement vector (in element co-ordinates) in part h, determine the horizontal (X^G) and vertical (Y^G) deflection at the mid-span of element 3 ($x = L^3/2$). The shape functions (applied to the frame element numbering system) are given below. Include your working. **[4 Marks]**

List of the shape functions for a frame element.			
$\psi_1(x) = \left(1 - \frac{x}{L}\right); \psi_2(x) = \frac{x}{L}$			
$N_1(x) = 1 - \frac{3x^2}{L^2} + \frac{2x^3}{L^3}$	$N_2(x) = \frac{x^3}{L^2} - \frac{2x^2}{L} + x$	$N_3(x) = \frac{3x^2}{L^2} - \frac{2x^3}{L^3}$	$N_4(x) = \frac{x^3}{L^2} - \frac{x^2}{L}$
		$u(x) = \psi_1(x)d_1 + \psi_2(x)d_4$ $v(x) = N d^e = \begin{pmatrix} N_1(x) & N_2(x) & N_3(x) & N_4(x) \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$	



All elements are FRAME elements

Elastic Modulus: $E_1 = E_2 = E_3 = 200.0$ GPa

Second Moment of Area: $I_1 = I_2 = I_3 = 4.5 \times 10^{-6}$ m⁴

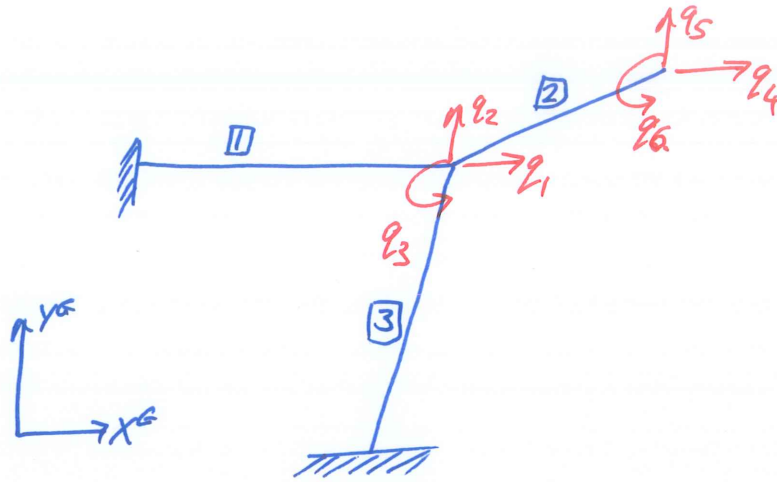
Cross-sectional Area: $A_1 = A_2 = A_3 = 3 \times 10^{-4}$ m²

Lengths: $L_1 = 1.50$ m

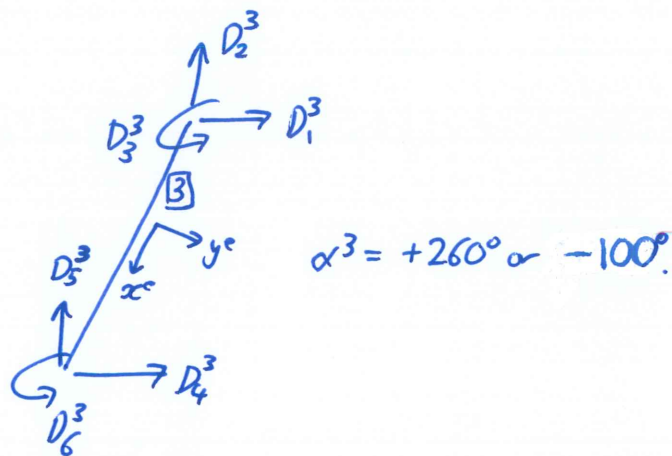
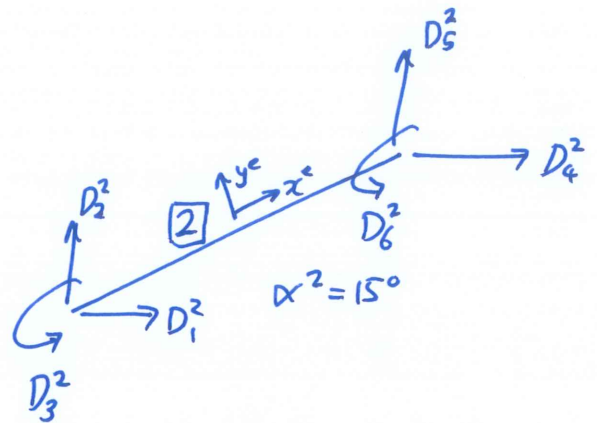
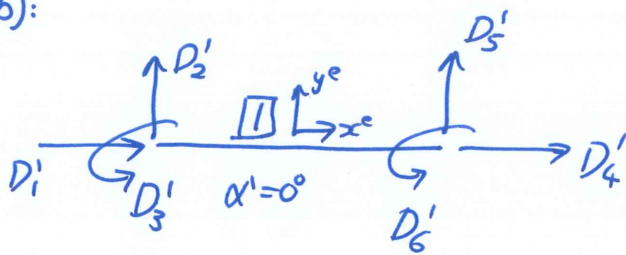
$L_2 = 2.00$ m

$L_3 = 2.50$ m

Q2a)



Q2b):



Q2c):

$$A^1 = \begin{matrix} & D_1' & D_2' & D_3' & D_4' & D_5' & D_6' \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$A^2 = I_{6 \times 6} = \begin{matrix} & D_1^2 & D_2^2 & D_3^2 & D_4^2 & D_5^2 & D_6^2 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$A^3 = \begin{matrix} & D_1^3 & D_2^3 & D_3^3 & D_4^3 & D_5^3 & D_6^3 \\ \begin{matrix} q_1 \\ q_2 \\ q_3 \\ q_4 \\ q_5 \\ q_6 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Element 1:

$$Q2d): f_{eq, uDL}^1 = \bar{w} \begin{Bmatrix} 0 \\ L/2 \\ (L)^2/12 \\ 0 \\ L/2 \\ -(L)^2/12 \end{Bmatrix}$$

$$\bar{w} = -(1.0 \times 10^3)(5.00) = -5 \times 10^3 \text{ N/m} = -5 \text{ kN/m}$$

↑
units kPa
or kN/m²

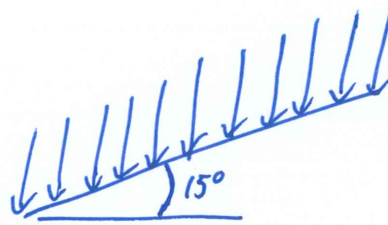
$$f_{eq, uDL}^1 = (-5000) \begin{Bmatrix} 0 \\ 1.5/2 \\ (1.5)^2/12 \\ 0 \\ 1.5/2 \\ -(1.5)^2/12 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ -3,750 \text{ N} \\ -937.5 \text{ Nm} \\ 0 \text{ N} \\ -3,750 \text{ N} \\ +937.5 \text{ Nm} \end{Bmatrix}$$

due to the fact that $\alpha' = 0^\circ$, $\Delta' = I_{6 \times 6}$ and $F_{eq, uDL}^1 = f_{eq, uDL}^1$

$$Q_{eq}^1 = \begin{Bmatrix} 0 \text{ N} \\ -3,750 \text{ N} \\ 937.5 \text{ Nm} \\ 0 \text{ N} \\ 0 \text{ N} \\ 0 \text{ N/m} \end{Bmatrix}$$

Element 2:

$$f_{eq, shear}^2 = (-4,829.6) \begin{Bmatrix} 0 \\ 2/2 \\ 2^2/12 \\ 0 \\ 2/2 \\ -2^2/12 \end{Bmatrix} = \begin{Bmatrix} 0 \text{ N} \\ -4,829.6 \text{ N} \\ -1,609.9 \text{ Nm} \\ 0 \text{ N} \\ -4,829.6 \text{ N} \\ +1,609.9 \text{ Nm} \end{Bmatrix}$$



Resolve downwards force into shear and axial components.

$$\bar{w} = -5000 \cos(15^\circ) = -4,829.6 \text{ N/m}$$

$$\bar{p} = -5000 \sin(15^\circ) = -1,294.1 \text{ N/m}$$

$$F_{eq, shear}^2 = (\Delta^2)^T f_{eq, shear}^2 = \begin{Bmatrix} 1,250 \text{ N} \\ -4,665 \text{ N} \\ -1,609 \text{ Nm} \\ 1,250 \text{ N} \\ -4,665 \text{ N} \\ 1,609 \text{ Nm} \end{Bmatrix}$$

$$f_{eq, axial}^2 = \bar{p} \begin{Bmatrix} L/2 \\ 0 \\ 0 \\ L/2 \\ 0 \\ 0 \end{Bmatrix} = (-1,294.1) \begin{Bmatrix} 2/2 \\ 0 \\ 0 \\ 2/2 \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} -1,294.1 \text{ N} \\ 0 \text{ N} \\ 0 \text{ Nm} \\ -1,294.1 \text{ N} \\ 0 \text{ N} \\ 0 \text{ Nm} \end{Bmatrix}$$

Q2d) - continued:

$$F_{eq,axial}^2 = (\Lambda^2)^T f_{eq,axial}^2 = \begin{Bmatrix} -1,250 N \\ -334.9 N \\ 0 Nm \\ -1,250 N \\ -334.9 N \\ 0 Nm \end{Bmatrix}$$

$$F_{eq,total}^2 = F_{eq,shear}^2 + F_{eq,axial}^2 = \begin{Bmatrix} 0 N \\ -5,000 N \\ -1,609.9 Nm \\ 0 N \\ -5,000 N \\ +1,609.9 Nm \end{Bmatrix}$$

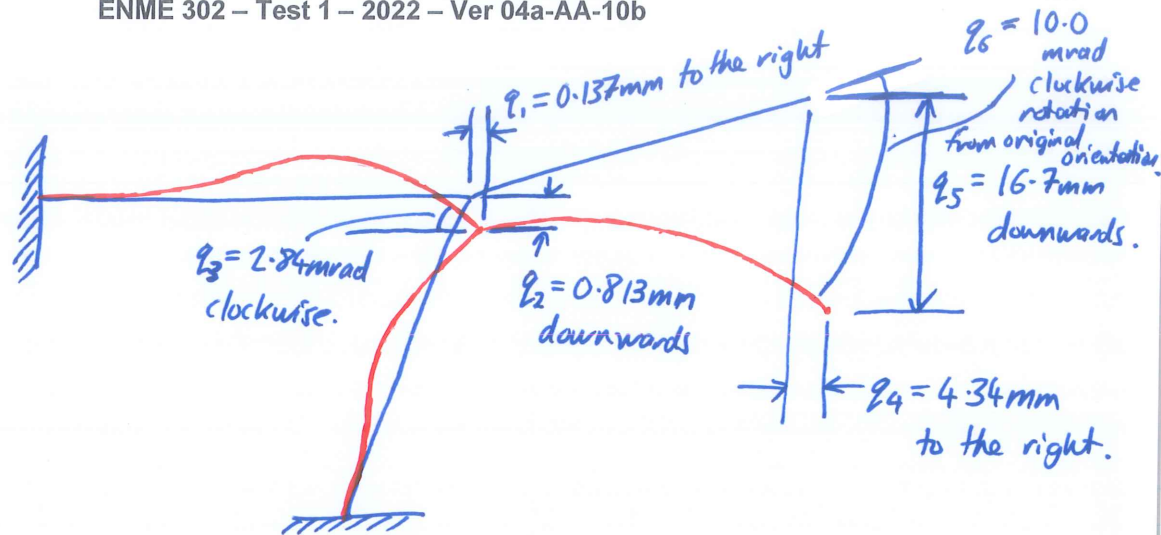
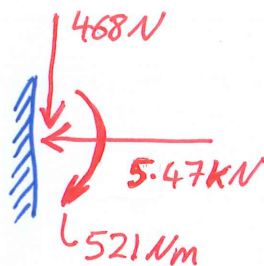
and as the assembly matrix for element [2] is an identity matrix,

$$Q_{eq,total}^2 = A^2 F_{eq,total}^2 = F_{eq,total}^2$$

$$Q_{total} = Q_{eq}^1 + Q_{eq,total}^2 = \begin{Bmatrix} 0 N \\ -3,750 N \\ 937.5 Nm \\ 0 N \\ 0 N \\ 0 Nm \end{Bmatrix} + \begin{Bmatrix} 0 N \\ -5,000 N \\ -1,609.9 Nm \\ 0 N \\ -5,000 N \\ +1,609.9 Nm \end{Bmatrix} = \begin{Bmatrix} 0 N \\ -8,750 N \\ -672.4 Nm \\ 0 N \\ -5,000 N \\ +1,609.9 Nm \end{Bmatrix}$$

Q2e)

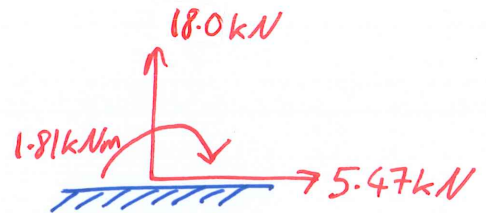
$$q = KG \setminus Q_{total} = \begin{Bmatrix} 0.0001 m \\ -0.0008 m \\ -0.0028 rad \\ 0.0043 m \\ -0.0167 m \\ -0.0100 rad \end{Bmatrix} = \begin{Bmatrix} 0.1367 mm \\ -0.8129 mm \\ -2.8414 mrad \\ 4.3436 mm \\ -16.6801 mm \\ -9.9964 mrad \end{Bmatrix}$$

Q2f):Q2g):

$$F_1(1:3) + (-F_2'(1:3))$$

$$= \begin{Bmatrix} -5,467.6 \text{ N} \\ -4,218.0 \text{ N} \\ -1,458.7 \text{ Nm} \end{Bmatrix} + \begin{Bmatrix} 0 \text{ N} \\ +3,750 \text{ N} \\ +937.5 \text{ Nm} \end{Bmatrix}$$

$$= \begin{Bmatrix} -5,467.6 \text{ N} \\ -468 \text{ N} \\ -521.2 \text{ Nm} \end{Bmatrix}$$



$$F^3(4:6) = \begin{Bmatrix} 5,467.6 \text{ N} \\ 17,968.0 \text{ N} \\ -1,807.6 \text{ Nm} \end{Bmatrix}$$

Q2h):

$$d^3 = \begin{Bmatrix} 0.7769 \text{ mm} \\ 0.2758 \text{ mm} \\ -2.8414 \text{ mrad} \\ 0 \text{ mm} \\ 0 \text{ mm} \\ 0 \text{ mrad} \end{Bmatrix}$$

$$\epsilon^3 = \frac{d_4^3 - d_1^3}{L^3} = \frac{(0) - (0.7769 \times 10^{-3})}{2.5}$$

$$\epsilon^3 = -3.1076 \times 10^{-4} \text{ } \epsilon$$

$$= -310 \text{ } \mu \epsilon$$

this strain is compressive
and has a magnitude of $310 \text{ } \mu \epsilon$.

Q2i): $\psi_1(x = \frac{L}{2}) = \psi_1(1.25) = (1 - \frac{1.25}{2.5}) = 0.5$

$$\psi_2(x = \frac{L}{2}) = \psi_2(1.25) = \frac{1.25}{2.5} = 0.5.$$

$$u(x) = \psi_1(x)d_1 + \psi_2(x)d_4$$

$$= (0.5)(0.7769) + (0.5)(0)$$

$u(x) = 0.38845 \text{ mm.}$

$$v(x) = N_1(x)d_2 + N_2(x)d_3 + N_3(x)d_5 + N_4(x)d_6$$

$$N_1(x = 1.25 \text{ m}) = 1 - \frac{3(1.25)^2}{(2.5)^2} + \frac{2(1.25)^3}{(2.5)^3} = 1 - 0.75 + 0.25$$

$$= 0.5.$$

$$N_2(x = 1.25 \text{ m}) = \frac{(1.25)^3}{(2.5)^2} - \frac{2(1.25)^2}{(2.5)} + (1.25) = 0.3125 - 1.25 + 1.25$$

$$= 0.3125.$$

$$v(x) = (0.5)(0.2758) + (0.3125)(-2.8414)$$

$v(x) = -0.7500 \text{ mm.}$

$$\delta X^G = u(x) \cos(\alpha^3) - v(x) \sin(\alpha^3)$$

$$= (0.38845) \cos(260^\circ) - (-0.75) \sin(260^\circ)$$

$$= -0.8061 \text{ mm.}$$

$$\delta Y^G = u(x) \sin(\alpha^3) + v(x) \cos(\alpha^3)$$

$$= (0.38845) \sin(260^\circ) + (-0.7500) \cos(260^\circ)$$

$$= -0.2523 \text{ mm.}$$

The horizontal X^G deflection of the mid-point of element [3] is 0.81 mm to the left. The vertical (Y^G) deflection of the mid-point of element [3] is 0.25 mm downwards.